· R-trees create regions in the cartesian plane in the shape of rectangles, while k-d trees split the plane along lines;
· While k-d trees alternate the dimension along which the split is done, R-tree doesn’t cycle through dimensions: rather, at each level the sub-rectangles created can partition their bounding box in any or even all dimensions at the same time;
· The bounding rectangles can overlap, both across different sub-trees and even with sibling’s bounding boxes, sharing the same parent. However, and this is crucial, no sub-rectangle extends outside its parent’s bounding box.
· Each internal node defines a so-called *bounding envelope*, that for R-trees is the smallest rectangle containing all the bounding envelopes of the node’s children.